ECHS FRAUD ANALYTICS

Pairwise Network Collusion Flagging

Beneficiary - Doctor - Hospital Relationship Analysis

CLASSIFICATION	RESTRICTED
REPORT PURPOSE	Boardroom representation, audit prioritisation and pairwise fraud-ring screening
INPUT EXTRACTS ANALYSED	4 network CSV extracts + hospital-name mapping workbook
ANALYTICAL TECHNIQUES	Pair frequency, percentile thresholds, Z-score validation and multi-pair flag convergence
PATTERN UNDER INVESTIGATION	Repeated beneficiary-doctor, doctor-hospital, beneficiary-hospital and full triad relationships
REPORT DATE	May 2026

Prepared for senior review - data-driven audit prioritisation, not direct enforcement action. Beneficiary names are anonymised, hospital IDs are replaced with hospital names or mapping labels, and every pairwise flag is positioned as an audit trigger requiring evidence validation.

Objective	How this report supports the project
Identify fraud patterns	Find repeated relationship structures across beneficiary, doctor and hospital dimensions.
Improve audit efficiency	Convert large extracts into ranked pairwise and combined audit queues.
Enable predictive decisions	Create explainable features for future risk routing without using automatic denial.

EXECUTIVE DASHBOARD - PAIRWISE FLAGGING ASSESSMENT

This report refines the pairwise network-collusion view into a clean, no-graph boardroom format. It separately examines beneficiary-to-doctor, doctor-to-beneficiary, doctor-to-hospital, hospital-to-doctor, beneficiary-to-hospital, hospital-to-beneficiary and combined full-triad patterns. The purpose is audit prioritisation, not allegation.

2,948 **Rows analysed** *relationship extract*	**1,856** **Beneficiaries** *anonymised*	**608** **Clean doctors** *normalised labels*	**199** **Named hospitals** *mapped labels*
1,171 **B-D pairs** *pairwise view*	**414** **D-H pairs** *claims view*	**1,220** **B-H pairs** *routing view*	**646** **Full triads** *combined view*

Fraud sub-pattern	Key pairwise signal	Flag threshold	Audit decision
Beneficiary-Doctor repeat	39 pairs repeated >= 3; max = 7	Same beneficiary + same doctor	Clinical continuity check
Doctor-Beneficiary concentration	Top doctor has 149 records across 94 beneficiaries	Doctor volume and beneficiary span	Speciality / roster validation
Doctor-Hospital concentration	P99 = 10.7 claims; max = 25	Doctor + named hospital	Provider relationship audit
Hospital-Doctor concentration	Hospital perspective checks top-doctor share and doctor spread	High claims or top-doctor share	Roster and empanelment check
Beneficiary-Hospital repeat	17 pairs repeated >= 5; max = 21	Same beneficiary + same hospital	Proximity / referral validation
Full triad convergence	16 triads repeated >= 3; max = 7	Beneficiary + doctor + hospital	Document-level validation

Boardroom conclusion: the current extracts are strong enough to create a pairwise and combined fraud-risk audit queue. They are not sufficient to allege collusion until dates, diagnosis, procedure, referral source, doctor speciality, beneficiary location and billing evidence are joined.

1. DATA SCOPE AND PAIRWISE FLAGGING METHODOLOGY

Each analytical view is treated as a relationship table. The same underlying data is examined from both directions because a beneficiary-centric repeat may look clinically normal, while the reverse doctor-centric or hospital-centric view may show concentration that needs audit explanation.

Analytical view	Source	Flag condition	Interpretation	Validation focus
Beneficiary -> Doctor	Sheet3	count >= 3; critical >= 5	Repeated doctor preference or channel	Clinical notes, speciality, past treatment
Doctor -> Beneficiary	Sheet3 + Sheet5	doctor records at P99; wide beneficiary span	High-volume doctor pattern	Roster, GP/specialist role, OPD schedule
Doctor -> Hospital	Sheet6	claims >= P95 or P99	Doctor-provider channel concentration	Doctor attachment, referrals, claim chronology
Hospital -> Doctor	Sheet6	high hospital claims or top-doctor share	Provider dependence on few doctors	Empanelment, hospital roster, specialty mix
Beneficiary -> Hospital	Sheet4	count >= 5; critical >= 10	Repeated provider routing	Proximity, chronic care, referral route
Hospital -> Beneficiary	Sheet4 + Sheet5	hospital rows, beneficiary spread, repeat beneficiaries	Provider-level beneficiary concentration	Catchment, treatment type, polyclinic route
Full triad	Sheet5	triad count >= 3; critical >= 5	Strongest combined relationship signal	Dates, ICD/procedure, bills, referral source

Metric	Population	Median	P95	P99	Max	Risk interpretation
Beneficiary-Doctor frequency	1,171	1	2.0	3.3	7	Repeated doctor relationship
Beneficiary-Hospital frequency	1,220	1	3.0	5.0	21	Repeated provider routing
Doctor-Hospital claims	414	1	5.0	10.7	25	Provider-channel concentration
Full triad frequency	646	1	2.0	4.0	7	Combined relationship repetition

Data quality check	Observed volume	Audit handling
Doctor not captured	1,515	Do not overstate doctor-level conclusions for these rows.
Hospital missing/invalid	1,173	Exclude from provider action until mapped.
Hospital names resolved by mapping	Hospital IDs replaced with names/mapping labels	Raw hospital IDs are not used in boardroom tables.

2. BENEFICIARY-DOCTOR AND DOCTOR-BENEFICIARY FLAGS

Beneficiary-doctor repetition is a direct continuity-of-care signal. It should be flagged, but not treated as fraud on its own. The reverse doctor-beneficiary perspective identifies high-volume clinicians and verifies whether the pattern is explainable by speciality, GP role, roster or reputation.

Metric	Current extract reading	Audit meaning
Repeated B-D pairs	39 pairs repeated >= 3	High-priority continuity review
Maximum B-D frequency	7 visits/records	Critical if not clinically explained
Top doctor concentration	149 records across 94 beneficiaries	Check GP/specialist role and roster

2A. Beneficiary -> Doctor repeated relationships

Beneficiary	Doctor label	Frequency	Doctor hospitals	Tier
BEN-0625	KAMAL BAGHI	7	1	CRITICAL
BEN-1082	PRAJYOT JAGTAP	5	1	CRITICAL
BEN-0684	R RAMESH	5	0	CRITICAL
BEN-0084	B P SINGH ORTHO	4	1	HIGH
BEN-0148	M R JAIN	4	1	HIGH
BEN-0166	SUBODH SINHA	4	1	HIGH
BEN-0171	VAIBHAV AGARWAL	4	1	HIGH
BEN-0713	K M BHANDARI	4	1	HIGH
BEN-1071	S RASTOGI GENERAL SURG	4	1	HIGH
BEN-0219	S M ROPLEKAR	4	0	HIGH

2B. Doctor -> Beneficiary concentration

Doctor label	Records	Beneficiaries	Hospitals	Raw variants	Audit lens
S M ROPLEKAR	149	94	2	15	High-volume GP/specialist check
S CHOPRA	30	26	1	1	High-volume GP/specialist check
JAYANT SARWATE	29	25	1	1	High-volume GP/specialist check
VAIBHAV AGARWAL	25	19	2	1	High-volume GP/specialist check
S GUPTA MD MEDICINE	19	16	1	1	Context validation
NIDHI GUPTA	19	15	1	2	Context validation
KULDEEP SINGH	17	11	1	6	Context validation
ANSHUMAAN V KAPOOR	16	15	1	1	Context validation

Balanced interpretation: a beneficiary may request the same doctor because the doctor understands the case history, is a suitable specialist, or is a trusted general physician. The flag becomes stronger only when it combines with unusual hospital routing, short timing gaps or billing abnormalities.

3. DOCTOR-HOSPITAL AND HOSPITAL-DOCTOR FLAGS

Doctor-hospital concentration is the operational bridge between clinical recommendation and provider billing. The doctor-to-hospital view ranks provider channels by claim count; the hospital-to-doctor view checks whether a provider is dependent on a few doctors or shows broad legitimate doctor coverage.

Metric	Current extract reading	Audit meaning
Doctor-hospital P95 / P99	5.0 / 10.7 claims	Pairs above P99 move to priority review
Maximum D-H claims	25 claims	Provider-channel concentration
Hospitals with broad doctor spread	Reviewed using doctor count and top-doctor share	Avoids over-flagging large providers

3A. Doctor -> Hospital claim concentration

Doctor label	Hospital name	Status	Claims	Tier
S CHOPRA	GETWELL HOSPITAL	INFERRED	25	CRITICAL
JAYANT SARWATE	Unknown - Manual Lookup Required	UNMATCHED	21	CRITICAL
S GUPTA MD MEDICINE	PRAYAG HOSPITAL	INFERRED	18	CRITICAL
VAIBHAV AGARWAL	Unknown - Manual Lookup Required	UNMATCHED	15	CRITICAL
DEEPAK TALWAR	METRO HOSPITAL, NOIDA	INFERRED	11	CRITICAL
KAMAL BAGHI	ABH HOSPITAL, FAZILKA	INFERRED	9	HIGH
M R JAIN	Unknown - Manual Lookup Required	UNMATCHED	9	HIGH
SIMRAN SINGH	AIMS HOSPITAL, RAIPUR	INFERRED	9	HIGH
ANURAG GARG	SUMITRA HOSPITAL	INFERRED	8	HIGH
MADHAV DHARME	Unknown - Manual Lookup Required	UNMATCHED	8	HIGH

3B. Hospital -> Doctor concentration

Hospital name	Claims	Doctors	Top doctor	Top doctor share	Risk lens
Unknown - Manual Lookup Required	208	112	JAYANT SARWATE	10.1%	Master-data priority
PRAYAG HOSPITAL	96	41	S GUPTA MD MEDICINE	18.8%	Broad doctor spread
GETWELL HOSPITAL	42	5	S CHOPRA	59.5%	Channel concentration
GRECIAN SUPER SPECIALITY HOSPITAL, BILASPUR	29	10	NIDHI GUPTA	24.1%	Broad doctor spread
AIMS HOSPITAL, RAIPUR	23	11	SIMRAN SINGH	39.1%	Broad doctor spread
METRO HOSPITAL, NOIDA	17	5	DEEPAK TALWAR	64.7%	Channel concentration
SGL SUPER SPECIALITY HOSPITAL, JALANDHAR	16	10	AJAY KUMAR	31.2%	Broad doctor spread
SDMC MSH HOSPITAL	15	12	ANUPAMA	20.0%	Broad doctor spread

Audit priority: doctor-hospital pairs above the 99th percentile need roster, empanelment, speciality and claim chronology review. A high share can be legitimate when the doctor is attached to the hospital or runs OPD there.

4. BENEFICIARY-HOSPITAL AND HOSPITAL-BENEFICIARY FLAGS

Beneficiary-hospital repetition captures repeated provider routing. This can indicate risk, but it can also be legitimate when the hospital is nearby, empanelled, specialised, or already managing a chronic condition. The hospital-to-beneficiary view helps distinguish provider concentration from ordinary catchment volume.

Metric	Current extract reading	Audit meaning
Repeated B-H pairs	17 pairs repeated >= 5	Proximity and continuity validation queue
Maximum B-H frequency	21 records	Critical unless route is justified
Named hospital perspective	199 provider labels reviewed	Separates catchment volume from repeat routing

4A. Beneficiary -> Hospital repeated routing

Beneficiary	Hospital name	Status	Frequency	Doctors observed	Tier
BEN-0258	Unknown - Manual Lookup Required	UNMATCHED	21	0	CRITICAL
BEN-1708	Unknown - Manual Lookup Required	UNMATCHED	9	2	HIGH
BEN-1587	Unknown - Manual Lookup Required	UNMATCHED	9	1	HIGH
BEN-0625	ABH HOSPITAL, FAZILKA	INFERRED	7	1	HIGH
BEN-0671	Unknown - Manual Lookup Required	UNMATCHED	6	3	HIGH
BEN-1082	SURANA SETHIA HOSPITAL, MUMBAI	INFERRED	6	2	HIGH
BEN-1238	Unknown - Manual Lookup Required	UNMATCHED	6	1	HIGH
BEN-1470	Unknown - Manual Lookup Required	UNMATCHED	6	1	HIGH
BEN-0015	ECHS POLYCLINIC - RANNI	POLYCLINIC	6	0	HIGH
BEN-0151	NANDA HOSPITAL	INFERRED	6	0	HIGH

4B. Hospital -> Beneficiary concentration

Hospital name	Records	Beneficiaries	Doctors	Repeat beneficiaries	Risk lens
Unknown - Manual Lookup Required	472	304	112	8	Master-data remediation
PRAYAG HOSPITAL	101	62	41	0	High catchment volume
GETWELL HOSPITAL	100	70	5	1	High catchment volume
SREE HOSPITAL, KERALA	100	91	0	0	High catchment volume
EMLC HOSPITAL	85	48	0	0	High catchment volume
NANDA HOSPITAL	65	27	0	2	Review repeat routing
GRECIAN SUPER SPECIALITY HOSPITAL, BILASPUR	47	29	10	1	High catchment volume
ECHS HOSPITAL	40	23	5	1	High catchment volume

Validation principle: a repeated beneficiary-hospital relationship should be downgraded where distance, referral route, chronic disease follow-up or continuity of care explains the pattern.

5. FULL TRIAD AND COMBINED MULTI-PAIR FLAGS

The full triad combines all three entities: beneficiary, doctor and hospital. This is the strongest screening view because it captures repeated routing through the same clinical and provider relationship. The combined flag below consolidates pairwise evidence into an audit queue.

Metric	Current extract reading	Audit meaning
Repeated full triads	16 triads repeated >= 3	Document-level validation queue
Maximum triad frequency	7 records	Critical if not explained by treatment chronology
Critical combined tier	12 triads	Immediate evidence pack review

5A. Combined beneficiary-doctor-hospital triad watchlist

Beneficiary	Doctor	Hospital	Triad freq	B-D freq	B-H freq	D-H claims	Risk tier
BEN-0625	KAMAL BAGHI	ABH HOSPITAL, FAZILKA	7	7	7	9	CRITICAL
BEN-1082	PRAJYOT JAGTAP	SURANA SETHIA HOSPITAL, MUMBAI	5	5	6	5	CRITICAL
BEN-0148	M R JAIN	Unknown - Manual Lookup Required	4	4	4	9	CRITICAL
BEN-1071	S RASTOGI GENERAL SURG	PRAYAG HOSPITAL	4	4	4	8	CRITICAL
BEN-0084	B P SINGH ORTHO	PRAYAG HOSPITAL	4	4	4	6	CRITICAL
BEN-0166	SUBODH SINHA	VENU EYE HOSPITAL	4	4	4	5	CRITICAL
BEN-0171	VAIBHAV AGARWAL	ECHS POLYCLINIC - KARAD	4	4	4	5	CRITICAL
BEN-1587	M R JAIN	Unknown - Manual Lookup Required	3	3	9	9	CRITICAL
BEN-1708	V MANIYA MD	Unknown - Manual Lookup Required	3	3	9	3	CRITICAL
BEN-1238	SHAILENDRA SHARMA	Unknown - Manual Lookup Required	3	3	6	3	CRITICAL

5B. Combined tier distribution

Combined tier	Triads	Boardroom interpretation
CRITICAL	12	Multiple pairwise triggers or critical triad repetition; immediate evidence pack review.
HIGH	80	Repeated triad or two strong pairwise triggers; targeted audit sampling.
MONITOR	88	One pairwise trigger; keep in watch queue until dates and claims are joined.
BASELINE	466	No major pairwise trigger in the current extract.

Important: the combined tier is a prioritisation index. It is designed to direct audit effort, not to conclude collusion. Provider action requires documentary confirmation.

6. INTERPRETATION AND CONCLUSION

A reputed-company presentation should show that the analytics is both powerful and balanced. The flags identify abnormal relationship patterns, while final decisions must be based on clinical, geographic, referral and billing validation. The report therefore separates audit prioritisation from enforcement action.

Beneficiary-Doctor pattern

Repeated contact with the same doctor can reflect continuity of care, specialist suitability, or beneficiary trust. It becomes higher risk when the same beneficiary, doctor and hospital recur together without a clear treatment chronology.

Doctor-Hospital pattern

A concentrated doctor-hospital channel may be normal when the doctor is attached to the hospital, runs OPD there, or is the relevant specialist. It needs review when the volume is unusually high relative to peers.

Beneficiary-Hospital pattern

Repeated visits to the same hospital may be explained by proximity, empanelment, chronic care, or post-operative follow-up. Distance and referral route should be checked before escalation.

Full triad pattern

The strongest watchlist cases are those where beneficiary, doctor and hospital repeat together and also show supporting pairwise flags. These should be sampled first for document-level review.

Priority	Recommended action	Purpose
1	Validate CRITICAL combined triads and P99 doctor-hospital channels	Focus audit effort on the highest convergence cases.
2	Resolve Unknown - Manual Lookup Required hospital labels before provider outreach	Prevent action against ambiguous or unmapped entities.
3	Add clinical, location and roster information to the review pack	Reduce false positives from proximity, chronic care or attached-doctor patterns.
4	Use reviewed outcomes as labels for future predictive routing	Improve the next version of the risk model using confirmed audit results.

Conclusion	Evidence from current extract	Business implication
C1. Pairwise fraud-risk signals are measurable	1,171 B-D pairs, 414 D-H pairs, 1,220 B-H pairs	Audit queues can be generated immediately.
C2. Full triad repetition is the strongest signal	16 triads repeated >= 3; maximum = 7	Prioritise evidence collection for repeated triads.
C3. Hospital names improve interpretability	Hospital IDs replaced with hospital names or mapping labels	Senior users can review without decoding raw IDs.
C4. Data gaps must be fixed before enforcement	1,515 doctor-missing rows and 1,173 hospital gaps	Avoid false positives caused by extraction or master-data issues.

Final executive statement: the pairwise and combined analysis demonstrates a practical fraud-ring screening approach for ECHS claims data. The output should be used for targeted audit prioritisation, followed by clinical, geographic, referral and billing-line validation.